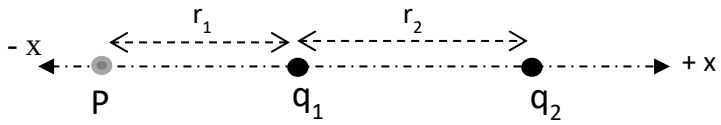
Physics 2 Problem Solving Video – Electric Field Part A

1) Two charged spheres with a net charge of $q_1 = 5$ nC positive and $q_2 = 15$ nC negative, using $r_1 = 0.4$ m and $r_2 = 0.6$ m. Determine the magnitude and direction of electric field at point P?

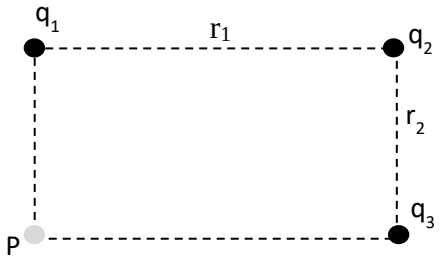


Answer: - 146 N/C

2) Shown in the figure below, three charged objects are placed at corners of a rectangle. The magnitude and type of charge is given below. Determine the magnitude and direction of the net electric field strength at point P. ($k = 9.0 \times 10^9$ Nm^2/C^2)

$q_1 = 2$ nC and is positive.
$q_2 = 30$ nC and is positive.
$q_3 = 5$ nC and is negative.

$r_1 = 0.5$ m
$r_2 = 0.3$ m



Net electric field is equal to $\Sigma\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$

Answer: $\left|\Sigma\vec{E}\right| = 918.3$ *N/C* $\quad \alpha = 105°$ *and* $\beta = 165°$